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Volta Industrial

Downtime & Maintenance Rescue: pull now, run, or expedite?

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$3.5M of production is wearing toward a stop, and the nightly report finds out after the shift.

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The stakes

A high-utilization run wore a cluster of lines toward failure

~90 critical lines

Across 8 plants, with rising vibration and temperature

$3.52M exposure

Downtime-at-risk today, with 160 open work orders

$22K per hour

Cost of an unplanned stop. Worst case: LINE-0004 at PLANT-03

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What Sam Ortiz needs

VP Manufacturing Operations, non-technical, decides per line

- Know which lines will stop before the shift ends, not after
- A clear play per line: pull now, run to shift end, or expedite the part
- The dollars behind it: downtime avoided vs. what the play costs
- Approve in one click, with an audit trail
- AI spend that stays predictable per plant across the fleet

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One governed data journey

Raw data to an approved work order, on one platform

Stage	Databricks capability	What it does for Volta
Ingest + transform	Lakeflow Declarative Pipelines	Telemetry, work orders, parts: raw to silver to gold; ai_classify on notes
Govern	Unity Catalog + metric view	One schema, one KPI definition (mv_line_risk), grants, lineage
Predict	MLflow + Model Serving	failure_recommender ranks the 3 plays by net $, batch and live
Ask	Genie	Plain-English questions become SQL over the governed gold tables
Serve + act	Lakebase + Databricks App	Low-latency queue; agent drafts the work order; approval writes back
Govern the AI	Unity AI Gateway	Rate limits, inference tables and usage tracking on every model call

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The model: what it predicts, how we trust it

XGBoost + Optuna, tracked in MLflow, @prod in Unity Catalog

What it predicts

- For each line and each play: downtime $ avoided, and the cost of the play
- Net value = avoided minus cost; the highest net wins
- The same model version scores the queue and answers live what-ifs

Why we trust it

- Trained on 35K historical decisions, no after-the-fact features
- 18.6% lower error than the historical-average baseline
- Calibrated: predicted vs. actual within 1.5–12% per risk bucket

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LINE-0004: the decision

Failure risk 0.87, replacement part not stocked locally

Play	Downtime $ avoided	Cost of the play	Net value
Pull now (recommended)	$65.5K	$15.0K	+$50.5K
Expedite parts and run	$19.4K	$7.1K	+$12.3K
Run to shift end	$0.9K	$80.0K	−$79.1K
What-if: part stocked locally → expedite	$38.8K	$4.0K	+$34.8K, still below pull now

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What the model changed

The decision logic is now learned, not hand-coded

Old hand-coded rule

- Assumed a planned pull costs a flat $40K
- Said "expedite" on 82 of 155 lines
- No way to test a what-if

Learned from Volta's history

- A planned pull really costs about $15K
- 53 high-risk lines flip to "pull now"
- Live what-ifs from the served model

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The demo

Live in the Volta Plant Floor app

- See it: the plant-floor view, a red cluster of lines trending to a stop
- Ask why: Genie answers in plain English over the governed data
- Get the play: the agent ranks pull / run / expedite in dollars, with a live what-if
- Act: approve; the work order writes back to Lakebase and the queue updates
- Governed: every AI call goes through Unity AI Gateway

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Governed, predictable AI

Spend is bounded, visible and attributable

Unity Catalog

One schema and metric view; the app reads only what it is granted

Unity AI Gateway

Rate limits act as a spend cap; every call is logged to inference tables

Cost controls

Serverless everywhere; Lakebase and the model endpoint scale to zero

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Design decisions and trade-offs

What I chose, and what it costs

Decisions

- Predict expected $, rank by net value
- Exclude post-action features (leakage)
- Judge by calibration, not R^2
- One model version for batch and live

Trade-offs

- A second model head to estimate cost
- Lower R^2 on paper (0.50)
- Lead-time rule is an explicit assumption
- Endpoint cost, so scale-to-zero

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Limitations and next steps

Where the prototype stops

- Exposure KPI (2h) and outcome history (4h) assume different stop lengths: reconcile to real MTTR
- History has no lead-time signal: capture it, then learn the rule instead of assuming it
- Built on synthetic data: accuracy on real telemetry is unproven
- First what-if after idle is slow (scale-to-zero): keep warm during shifts

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Next step: a 30-day pilot at PLANT-03

Prove it on real data before scaling to the fleet

- Scope: PLANT-03 lines, real telemetry and maintenance history
- Success: unplanned stops avoided, $ saved vs. exposure, approval rate
- Measure the model against what the plant would have done anyway
- Owners: Volta maintenance lead + Databricks field team
- Decision at day 30: expand to the 8-plant fleet

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